# CortexCall

EEG Motor Imagery Classification — Kickoff

*Reading LEFT vs. RIGHT hand imagination straight from brain signals — no movement required.*

Team Planning Session · Week 1

# The Big Picture

What we're building and how the whole pipeline fits together

**Goal:** Classify whether someone is imagining LEFT or RIGHT hand movement — using only their EEG signal, ~2000 recorded trials.


*We are here — EDA decides what the rest of the pipeline is built on.*

**Typical channels:** FZ, C3, CZ, C4 (more possible)  **Target users:** research team + future assistive-tech demo

# Understanding the Dataset

~2000 Excel files. Here's what's actually inside one of them.

*time ↓ channels →*

| | FZ | C3 | CZ | C4 |
|---|---|---|---|---|
| t1 | −0.93 | −0.04 | −0.13 | 0.51 |
| t2 | 0.34 | 0.29 | −0.99 | 0.69 |
| t3 | 0.40 | −0.12 | 0.36 | 0.24 |
| t4 | −0.45 | 0.86 | −0.06 | −0.18 |

**label for whole file: LEFT or RIGHT**

- **Columns = channels**

  Each column is one electrode (FZ, C3, CZ, C4...)

- **Rows = time samples**

  Recorded at 128–256 Hz, over 3–8 seconds

- **One label per file**

  The whole trial is LEFT or RIGHT — not per-row

**Watch this:** we must keep a clean filename → label mapping (CSV or in the filename itself) across all 2000 files. If this breaks, everything after it is compromised.

# The Signal We're Hunting

Everything else in this project only matters if this shows up.

**8–12 Hz**

Mu rhythm

**13–30 Hz**

Beta rhythm

**C3 / C4**

Motor cortex channels

**ERD (Event-Related Desynchronization):** power drops in these bands over C3/C4 when someone imagines moving — that dip is the “tell” we're looking for.

**Why it's the gate, not just a chart:** if PSD plots show no LEFT vs. RIGHT difference here, no model — classic or deep learning — will fix that. We check this before investing in modeling.

# This Week: EDA Only

All four of us, working in parallel, on the same raw data.

**Week 1**

Focus

### No sequencing needed within EDA

All 4 people can start immediately — everyone reads the same raw files.

### Different question each

We split by question, not by person, so nobody duplicates work.

### One shared conclusion

Findings converge into a single summary that Processing will depend on next.

**By Friday:** we should know whether the data is usable, and have the exact filter band, rejection rule, and epoch window Processing will use next week.

# Split by Question, Not by Person

Four notebooks, four different questions — not four copies of the same analysis.

| | |
|---|---|
| **Person A** | Class balance, sampling-rate consistency, missing values & NaNs |
| **Person B** | Time-domain plots — LEFT vs. RIGHT, raw vs. filtered, on C3/C4 |
| **Person C** | PSD + ERD analysis on C3/C4 — the “does this data even work” question |
| **Person D** | Artifact & outlier statistics — how many trials would we reject, and why |

# The Risk We Need to Avoid

If everyone cleans the raw data differently, our conclusions stop being comparable.

- Person A drops NaN rows one way
- Person B resamples the signal differently
- Person C uses different channel names

**Result:** everyone's "conclusions" are actually about different data — comparing apples answering different questions on different apples. We can't converge on one config from that.

**The fix is on the next slide — one shared loader, used by everyone.**

# The Fix: One Shared Loader

The 80/20 rule: most work stays in notebooks — the reusable 20% graduates to a script.

### 80% — Notebooks

- Plotting, PSD, exploration
- “Let me try X and see” experiments
- Personal, one-off, fast iteration
- Stays individual — not imported by others

### 20% — data_loader.py

- Standardizes channel names/order
- Attaches LEFT/RIGHT label
- Verifies sampling rate — flags mismatches
- No filtering yet — that's next week's decision
- Everyone imports it, nobody copy-pastes it

# Shared vs. Individual — At a Glance

| Frozen & Shared | Individual per Notebook |
|---|---|
| File loading logic | Specific plots & analysis |
| Column / channel naming | Exploratory filtering (just to look, not applied) |
| Label attachment (filename → LEFT/RIGHT) | Outlier / rejection proposals (findings, not decisions) |
| Sampling-rate & length verification | Interpretation and conclusions |

*Rule of thumb: will more than one person import this exact logic? If yes → it belongs on the left.*

# Project Structure

Folders mirror the pipeline stages, not the people.

```
cortex-call/
    data/
        (gitignored — raw + processed, local only)
    src/
        data_loader.py, config.py — shared, imported code
    notebooks/
        eda_*.ipynb + eda_summary.md — individual + converged
    models/
        (gitignored — trained weights)
    app/
        Streamlit demo (deployment stage)
    reports/
        final report, slides, figures
```

### `src/config.py`

The one file every stage depends on — worth guarding with review.

### `notebooks/eda_summary.md`

Converged conclusions — feeds config.py next phase.

### `data/ & models/`

Gitignored, but the folders exist with a note on where to download from.

# Team Workflow Rules

Lightweight, not enterprise — just enough to catch the mistakes that matter here.

### No direct pushes to main

main stays stable at all times

### One branch per task

e.g. eda-psd, eda-class-balance, csp-svm-baseline

### PR + 1 review before merge

Team lead reviews, especially anything touching config.py

### Flag preprocessing changes in the PR

“Changed epoch window from X to Y” — the thing most likely to silently break comparability

# Why Subject-Aware Splits Matter

A detail worth getting right before modeling starts.

### ✗ Random split

Subject 7's trials land in both train AND test. The model partly learns "what Subject 7's brain looks like," not LEFT vs. RIGHT. Accuracy looks great — falsely.

### ✓ Subject-aware split

Train on Subjects 1–16, test only on Subjects 17–20 — people the model has never seen. An honest estimate of real-world performance.

**Action item for EDA:** check whether our ~2000 files carry a subject ID anywhere — filename, metadata sheet, or folder structure. This decides which CV strategy we can actually use.

# Repo Setup

What we're calling it, and what stays out of git.

## CortexCall

Public repo · no license yet

*EEG-based motor imagery classification (LEFT vs. RIGHT hand) using signal processing, CSP+SVM baseline, and EEGNet deep learning, deployed as a Streamlit demo.*

`cortex-call  (lowercase-hyphenated is the safe convention across OSes)`

### .gitignore essentials

```
# Data
data/
*.xlsx
*.csv

# Models
*.pt
*.pkl
*.joblib
*.onnx

# Python
__pycache__/
venv/ .venv/
.ipynb_checkpoints/

# OS / Editor
.DS_Store
.vscode/
```

# Roadmap

Where we're headed after this week

- ✓ Repo & structure set up
- 2 Shared data_loader.py
- 3 EDA — 4 parallel notebooks
- 4 Converge into eda_summary.md → config.py
- 5 Preprocessing pipeline
- 6 CSP + SVM baseline
- 7 EEGNet model
- 8 Streamlit demo